

# PARTHASARATHY RAMARAJ

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Engineering Leader and Hands-On AI Engineer — 20+ Years Building Software, 12+ Leading Teams | Production AI Agents at JPMC

• Series A Interim CTO • Consumer-Scale Systems at NYT | Tech Blog: [rpsarathy.hashnode.dev](https://rpsarathy.hashnode.dev)

## PROFESSIONAL SUMMARY

Engineering leader and hands-on engineer with 20+ years building software and **12+ years leading teams**. Vice President, Sr Lead Software Engineer at J.P. Morgan Chase, writing production Java and Python daily on a firmwide Credit Risk platform and building production AI agents (LangGraph, MCP, RAG). As Series A interim CTO at Sidekick, grew engineering from **10 to 30+** and rebuilt the platform; at The New York Times, ran an **8,000 RPS** decision engine through peak election traffic. Suited to Engineering Manager, Staff and Principal Engineer, and Architect roles.

## CORE SKILLS

**AI-Native Platform, Agents & Evaluation: Built and shipped production AI agents at JPMC:** LangGraph multi-agent Text-to-SQL agent with MCP tooling on a RAG framework (15% ticket reduction); AI agent for production log analysis and incident triage; custom Claude Code subagents and reusable **SKILL.md library** as team standards; evaluation harness (traces, token cost, latency, offline replay, regression gates); LangChain, LlamaIndex; Anthropic Claude API; advisory experience with open-weight LLMs (Kimi, Llama, GLM) via Ollama at ZYRO Hive

**Distributed Systems, Scale & Modernization: 8,000 RPS decision-engine API at NYT** (peak election traffic); **Kafka streaming 3M+ records/day at Thrivent**; JPMC end-of-day batch re-engineered **12h to 6h**; Sidekick full re-platforming on micro-frontend, API-first, Kafka pub/sub; event-driven architecture; concurrency-tuned components; back-pressure, retry, and idempotency; SLOs and SLAs

**Languages: Java, Python & SQL: Java** (Spring, Spring Boot, concurrency, GC tuning) production-daily at JPMC and historically at Markit; **Python** (LangGraph tooling) daily at JPMC; **SQL** / PostgreSQL, Oracle, Cassandra, MongoDB, BigQuery, Neo4j; Go; basic working knowledge of React / TypeScript / Node.js

**API Design & Data Platform Architecture: Reusable GraphQL API design framework at NYT adopted across teams** (+20% consumer usage); REST / gRPC / GraphQL production APIs; OAuth 2.0 / JWT API security; API-first microservices; **Neo4j knowledge graphs and GraphRAG at ZYRO Hive**; ChromaDB vector database; RAG and embeddings

**Cloud, Kubernetes & Delivery: AWS 10+ years** (ECS, Lambda, EC2, S3, RDS, EMR, ALB / NLB, VPC, IAM, CloudWatch, serverless, multi-region resilience); Terraform; GCP (GKE, Cloud Run, BigQuery); Docker and **Kubernetes**; CI/CD (Jenkins, GitHub Actions); Sidekick AWS migration (**20% lower cloud cost**); ELK and Grafana observability

**Regulated-Data Compliance & Security:** JPMC firmwide Credit Risk (MTM stress, counterparty trades) under enterprise security and compliance guardrails; **GDPR/CCPA-compliant data governance** (identity, access, policy controls) at Sidekick, Thrivent, and NYT; Markit sell-side platforms under strict availability SLAs; secure coding, peer review, automated testing

## PROFESSIONAL EXPERIENCE

**J.P. Morgan Chase — Jersey City, NJ**

**Apr 2024 – Present**

*Vice President, Sr Lead Software Engineer*

- Senior IC and technical lead on a firmwide Credit Risk platform (MTM stress, counterparty trades): **writes and reviews production Java and Python code daily**, drives architectural design reviews and RFCs.
- Architected and shipped a **LangGraph multi-agent Text-to-SQL agent** with MCP-based tooling on a RAG framework, cutting engineering support tickets **15%** through retrieval grounding and validation.
- Co-developed an AI agent for production log analysis and incident triage: automated log-pattern extraction, root-cause hypotheses, and remediation suggestions under enterprise security and compliance guardrails.
- Instrumented the evaluation harness and observability for the agent stack: traces, token cost, latency, offline replay, and regression gates.
- Rolled out Claude Code CLI, MCP tooling, custom subagents, and a reusable **SKILL.md library** as team-wide AI-native engineering defaults under enterprise security and compliance guardrails.
- **Re-engineered the end-of-day risk batch from 12 hours to 6 hours** through parallel compute, serverless, and pipeline optimization, cutting infrastructure cost 20%.

**ZYRO Hive (Non-Profit Pre-Seed) — Remote**

**Oct 2023 – Present**

*Technical Adviser — AI & Knowledge Graph Architecture*

- Architected **LLM integration with knowledge graphs (Neo4j)**, enabling intelligent decision-support and symmetrical semantic search capabilities for complex investment-related datasets.
- Embedded participatory data stewardship principles into technical strategy, improving cross-functional collaboration and project outcomes by **30% over three quarters**.

**Thrivent Financial Services — Remote**

**Sep 2022 – Oct 2023**

*Senior Architect (Engineering Leader)*

- Engineered real-time **Kafka data streams processing 3M+ records/day** for analytics, segmentation, and decisioning on an event-driven architecture.
- Implemented an API-first microservices platform for downstream services (**+30% operational efficiency**), a shared backend other teams build against.

- Modernized CI/CD on Docker and Kubernetes and set API-first integration standards and design reviews across four scrum teams.

#### **Sidekick (formerly Avenue 8 — Series A PropTech) — Remote**

**Jun 2021 – Jun 2022**

##### *Engineering Director*

- Set technical direction as Series A interim CTO: architected a full re-platforming from Avenue 8 to Sidekick on a micro-frontend, API-first architecture with Kafka pub/sub, cutting deploy time **30%**.
- Drove the AWS migration with **GDPR/CCPA-compliant data governance** (identity, access, policy controls), cutting cloud cost 20% within 6 months.
- Stood up SRE and observability practices from scratch with ELK and Grafana: resilience metrics, monitoring, and alerting.
- Partnered directly with cofounders on technical roadmap, delivery cadence, and hiring across product, platform, and growth engineering.

#### **The New York Times — New York, NY**

**Jul 2015 – Jun 2021**

##### *Senior Software Engineer/Development Manager*

- Designed and operated a real-time decision-engine API sustaining **8,000 RPS under low-latency SLAs** through peak election traffic, serving multiple downstream frontends.
- Built a **reusable GraphQL API design framework adopted across teams**, raising consumer usage 20%.
- Embedded regulatory and compliance engineering into the SDLC, with security, privacy, and audit checkpoints wired into the CI pipeline.
- Migrated MarTech applications to GCP and AWS with monitoring and auto-healing patterns for consumer-scale traffic.

#### **Citi — Consulting Engagement**

**May 2014 – Jun 2015**

##### *Technical Lead / Architect (Consultant)*

- Consulting engagement as technical lead and architect: established an onsite and offshore engineering team, delivered architectural solutions to complex application problems, resolved production memory-leak and performance issues, and set up new development, SIT, and UAT environments.

#### **Markit (now S&P Global / IHS Markit) — New York, NY**

**Mar 2011 – May 2014**

##### *Lead Engineer | Vice President*

- Delivered low-latency, high-availability distributed **Java platforms under strict SLAs** for sell-side clients: resilient services, concurrency-tuned components, and secure data layers.
- Designed reference-data and post-trade services with reconciliation-first design, data-integrity guarantees, and multi-region availability.
- Tuned JVM performance and throughput on platforms handling market and reference data: GC profiling, memory tuning, and thread-model optimization under production load.

#### **Consulting Engagements (Multiple Clients) — United States**

**Jul 2002 – Mar 2011**

##### *Senior Consultant | Software Engineer*

- Delivered enterprise Java, SQL, and distributed-systems solutions for financial-services clients with full-lifecycle production ownership under enterprise SLAs.
- Served as technical lead on long-term Fortune 500 engagements: mentored engineers, ran design reviews, and shipped performance-critical distributed services.

## **EDUCATION**

**Bachelor of Engineering** — Bharathiar University, India | 2002